Locally Certified Worst-Group Reliability under Covariate Shift: When Subgroup Alarms Are Evidence

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Abstract

Worst-group audits are meant to answer an operational question: is there a subgroup for which a deployed model is unreliable enough to require action? Under covariate shift, the usual answer is to importance-weight labeled source data to represent the target population, then report the subgroup-bin cell with the largest estimated residual. This paper shows that such alarms can be statistically misleading. The relevant question is not only whether the shifted estimand is transportable, but whether the selected cell is locally supported by enough weighted labels to certify the alarm.

We develop a compact inference framework for worst-group reliability under covariate shift. The first result is an identifiability boundary: if a target subgroup-bin cell has no source support, its target reliability cannot be certified from labeled source data and unlabeled target covariates, even when covariate shift is otherwise correct. The main result gives simultaneous confidence certificates for a finite family of subgroup-bin cells. The radius is governed by cellwise effective sample size, not global effective sample size. The same theorem covers cross-fitted learned density ratios by adding an explicit local nuisance term, and separates conditional-on-covariates guarantees from population-target guarantees.

Experiments are chosen to test the statistical claim directly. In a calibrated covariate-shift benchmark, naive worst-cell audits and global-ESS gates produce frequent false alarms, while local simultaneous certificates control them and retain power when a planted subgroup failure has adequate overlap. A learned-weight and adaptive-search study shows how density-ratio error and subgroup search add uncertainty beyond oracle local ESS. Two real-data deployment-style audits, Adult Census and ACSIncome, show the practical consequence: large naive subgroup alarms would trigger escalation, but local certificates correctly report insufficient evidence when the alarm-driving cells are thinly supported. The main lesson is simple: under covariate shift, worst-group alarms should be reported only when they are locally supported and statistically certified.

1 Introduction

Consider a validation team auditing a risk model before deployment in a new population. The average calibration error looks acceptable. The global effective sample size after importance weighting is large enough that the audit seems usable. But the dashboard highlights one subgroup-bin cell as the worst: perhaps high-score patients in one hospital service line, or high-confidence decisions for a demographic intersection in a public-service model. The estimated gap is large enough to trigger a real action: subgroup investigation, recalibration, temporary suppression of the score, or model rollback.

What should the team conclude? There are two very different possibilities. The alarm may be a real subgroup failure that the aggregate metric hides. Or it may be an artifact: the target population puts mass on that subgroup-bin cell, but the weighted source audit set contains only a few effective labels there. In the first case, ignoring the alarm is dangerous. In the second, acting on it is also dangerous, because the audit has mistaken local overlap collapse for model failure.

This paper studies that distinction under covariate shift. We observe labeled source data from P, target covariates from Q_X, and assume

$$P(Y \mid X) = Q(Y \mid X), \qquad P_X \neq Q_X.$$

Importance weighting by $w(x) = dQ_X/dP_X(x)$ is the standard way to transport source labels to the target population. That step answers one question: what target moment should be estimated? It does not answer the audit question: is the reported worst subgroup alarm supported by enough local evidence to be trusted?

The gap between these questions is the central issue. A global shifted-risk estimate averages over the full weighted source sample. A worst-group audit instead takes a maximum over many subgroup-bin cells. The selected cell is often chosen precisely because its estimate is extreme, and extreme local estimates are most likely to occur where support is weak. Thus a global effective sample size can be reassuring while the alarm-driving cell is effectively supported by one or two weighted labels.

For a subgroup-bin cell c, let $A_c(X_i)$ indicate whether source point i falls in that cell. The relevant information scale is the local effective sample size

$$n_{\mathrm{eff}\,c} = \frac{\left(\sum_i w_i A_c(X_i)\right)^2}{\sum_i w_i^2 A_c(X_i)}.$$

This is the ordinary weighted ESS, but computed only inside the cell that may be reported as the worst. It has a direct interpretation: how many effectively independent weighted labels support the local alarm? If this number is small, a large point estimate is not yet evidence.

The practical conclusion is simple. A covariate-shift audit should not report only

$$\max_{c \in \mathcal{C}} |\widehat{r}_c| > \tau,$$

where τ is a tolerance and $\widehat{r}_c$ is a weighted cell residual. It should report a certified excess after subtracting a simultaneous local uncertainty radius. If the point estimate is large but the certified excess is zero, the right conclusion is not “no failure.” It is “insufficient evidence.”

1.1 What Existing Tools Do Not Decide

Several mature literatures touch this problem, but each answers a different question.

- The literature on **covariate shift and importance weighting** explains how to transport shifted risks or fixed target moments (Sugiyama et al., 2007; Cortes et al., 2010; Polo and Vicente, 2023). It does not by itself say whether the selected worst subgroup-bin alarm has enough local weighted labels to be trusted.
- The literature on **calibration, conformal prediction, and uncertainty under shift** provides shift-aware validity guarantees for scores or prediction sets (Guo et al., 2017; Park et al., 2020; Tibshirani et al., 2019; Yang et al., 2024). The object here is different: a maximum subgroup-bin residual after weighting, search, and local overlap variation.

- The literature on **multicalibration and subgroup auditing** explains why aggregate metrics can hide subgroup failures (Hébert-Johnson et al., 2018; Kearns et al., 2018; Wu et al., 2024; Zhang et al., 2024). Our question is how to distinguish a true subgroup failure from an unsupported shifted subgroup alarm.
- The literature on **slice discovery and post-selection inference** shows that adaptive search invalidates naive validation unless inference accounts for search (Chung et al., 2019; Eyuboglu et al., 2022; Berk et al., 2013; Dwork et al., 2015; Angelopoulos et al., 2021). We add the covariate-shift ingredients: local ESS and learned density-ratio nuisance inside the selected cell.

The paper is therefore not primarily a new density-ratio estimator, a new subgroup discovery algorithm, or a new calibration metric. It is an inference framework for deciding when a shifted worst-group alarm is evidence. The key object is local support inside the reported subgroup-bin cell.

1.2 Contributions

The through-line is:

> under covariate shift, worst-group alarms are trustworthy only when they are locally supported and statistically certified.

We make this claim precise in four steps.

- We formulate worst-group reliability auditing as localized weighted moment inference over subgroup-bin cells.
- We prove an identifiability boundary: without source support in the target cell, no method using source labels and target covariates can uniformly certify target reliability there.
- We give a local certificate theorem for finite audit families. The confidence radius is controlled by cellwise ESS, extends to cross-fitted learned weights through a cellwise nuisance term, and separates conditional finite-sample claims from population-target claims.
- We show empirically that the distinction changes decisions: naive and global-ESS-gated audits produce unsupported subgroup alarms, while local certificates refuse artifacts, retain power when overlap is adequate, and change decisions in Adult Census and ACS public-service audits.

2 Setting and Estimands

The model score $s : \mathcal{X} \to [0, 1]$ is fixed before the audit. We observe labeled source data $(X_i, Y_i)_{i=1}^n \sim P$ and target covariates $(X'_j)_{j=1}^m \sim Q_X$. Labels are binary, $Y \in \{0, 1\}$. The target labels may be unavailable at audit time.

Assumption 1 (Covariate shift and overlap)**.** The conditional label law is stable, $P(Y \mid X) = Q(Y \mid X)$, and $Q_X \ll P_X$. The density ratio

$$w(x) = \frac{dQ_X}{dP_X}(x)$$

is finite P_X-almost surely.

Let $\mathcal{G}$ be a finite family of subgroup indicators and let $\Pi = \{I_1, \ldots, I_B\}$ be a partition of the score range. A cell $c = (b, g)$ is active when

$$A_c(x) = \mathbf{1}\{s(x) \in I_b\} g(x) = 1.$$

The finite audited family is $\mathcal{C}$, with $M = |\mathcal{C}|$.

For cell c, the target mass and target residual mass are

$$\mu_c^Q = \mathbb{E}_Q[A_c(X)], \qquad \theta_c^Q = \mathbb{E}_Q[(Y - s(X))A_c(X)].$$

When $\mu_c^Q > 0$, the target cellwise residual is

$$r_c^Q = \mathbb{E}_Q[Y - s(X) \mid A_c(X) = 1] = \frac{\theta_c^Q}{\mu_c^Q}.$$

This is the quantity behind a subgroup-bin reliability alarm. If $r_c^Q > 0$, outcomes are larger than the score in that cell; if $r_c^Q < 0$, outcomes are smaller.

Under assumption 1, target moments are transported by importance weighting:

$$\mu_c^Q = \mathbb{E}_P[w(X)A_c(X)], \qquad \theta_c^Q = \mathbb{E}_P[w(X)(Y - s(X))A_c(X)].$$

The self-normalized estimator of r_c^Q is

$$\widehat{r}_c(w) = \frac{\sum_i w(X_i)A_c(X_i)(Y_i - s(X_i))}{\sum_i w(X_i)A_c(X_i)}, \tag{2.1}$$

whenever the denominator is positive. Its local effective sample size is

$$n_{\mathrm{eff}\,c}(w) = \frac{\left(\sum_i w(X_i)A_c(X_i)\right)^2}{\sum_i w(X_i)^2 A_c(X_i)}. \tag{2.2}$$

The global effective sample size,

$$n_{\mathrm{eff}\,\mathrm{global}}(w) = \frac{(\sum_i w(X_i))^2}{\sum_i w(X_i)^2},$$

answers a different question: how much information remains in the full weighted sample. Worst-group auditing asks whether the selected local cell is supported.

Given a tolerance $\tau > 0$, a certified audit should not report only the largest point estimate. It should report a lower confidence bound on the excess gap:

$$\mathrm{excess}_c = [|\widehat{r}_c| - B_c - \tau]_+ ,$$

where B_c is a simultaneous uncertainty radius. A certified failure requires positive excess. A large $|\widehat{r}_c|$ with zero certified excess is not a negative result; it is insufficient evidence.

3 Theory: From Local Support to Certified Alarms

The theory has one main message. Transport identifies the target residual, but local support determines whether the corresponding alarm can be certified.

3.1 The Identifiability Boundary

The first result explains why local overlap is not merely a numerical diagnostic. If the target cell has no source support, the target residual in that cell is not identified by source labels and target covariates.

Proposition 1 (No local overlap, no certification)**.** *Let $A \subseteq \mathcal{X}$ be a subgroup-bin cell with $Q_X(A) = \pi > 0$ and $P_X(A) = 0$. For any $\Delta \in (0, 1)$, there exist two covariate-shift worlds (P_0, Q_0) and (P_1, Q_1) such that the observable law of the labeled source sample and unlabeled target covariates is identical in the two worlds, but*

$$r_A^{Q_1} - r_A^{Q_0} = \Delta.$$

Consequently, no audit using only those observations can uniformly certify target reliability in A.

The point is simple. Target covariates can reveal that a cell exists in deployment, but if source labels never appear there, the conditional label law inside that cell is unconstrained by the audit data. A finite-sample version is immediate: when $P_X(A) = O(1/n)$, there is constant probability that the source audit set contains no label from A. Honest inference must remain wide on that event.

3.2 The Main Certificate

Now suppose the cell has observed support. Write

$$Z_i = Y_i - s(X_i), \qquad m_i = \mathbb{E}[Z_i \mid X_i].$$

For any nonnegative audit weights v_i, define

$$\widehat{r}_c(v) = \frac{\sum_i v_i A_c(X_i) Z_i}{\sum_i v_i A_c(X_i)}$$

and

$$\widehat{n_{\text{eff}}}_c(v) = \frac{\left(\sum_i v_i A_c(X_i)\right)^2}{\sum_i v_i^2 A_c(X_i)}.$$

The conditional target of this weighted mean is

$$\bar{r}_c(v) = \frac{\sum_i v_i A_c(X_i) m_i}{\sum_i v_i A_c(X_i)}.$$

This is the right finite-sample object when the source covariates and target covariates have been observed. Population certification adds one more transport term below.

Assumption 2 (Label-independent audit weights)**.** Conditional on the source and target covariates and on any randomness used to fit the weights, the audit weights v_i are fixed and the labels Y_i are independent. This holds for oracle weights $v_i = w(X_i)$. It also holds for learned weights $v_i = \widehat{w}_i$ when the weight assigned to source point i is trained without using Y_i, for example by source-side cross-fitting.

Learned weights also require a local nuisance bound. For each cell, let

$$S_c = \sum_i w(X_i) A_c(X_i), \qquad \Delta_c = \sum_i |\widehat{w}_i - w(X_i)| A_c(X_i).$$

Assume a conservative envelope ρ_c satisfies

$$\rho_c \geq \frac{2\Delta_c}{S_c - \Delta_c}, \qquad S_c > \Delta_c. \tag{3.1}$$

For oracle weights, set $\rho_c = 0$.

Theorem 1 (Locally certified subgroup-bin inference)**.** *Assume assumption 2. Let $M = |\mathcal{C}|$, $Y_i \in \{0,1\}$, $s(X_i) \in [0,1]$, and suppose all audited denominators are positive. Then, conditional on the covariates and fitted weights,*

$$\mathbb{P}\left(\forall c \in \mathcal{C} : |\widehat{r}_c(v) - \bar{r}_c(v)| \leq \sqrt{\frac{\log(2M/\delta_1)}{2\widehat{n_{\mathrm{eff}}}_c(v)}}\right) \geq 1 - \delta_1.$$

If $v_i = \widehat{w}_i$ are learned weights satisfying Equation (3.1), then on the same event,

$$\forall c \in \mathcal{C} : |\widehat{r}_c(\widehat{w}) - \bar{r}_c(w)| \leq \sqrt{\frac{\log(2M/\delta_1)}{2\widehat{n_{\mathrm{eff}}}_c(\widehat{w})}} \; + \; \rho_c.$$

Finally, suppose $0 \leq w(X) \leq W$. Let

$$\epsilon_n(\delta_2) = 2W\sqrt{\frac{\log(4M/\delta_2)}{2n}}, \qquad \gamma_c(\delta_2) = \frac{2\epsilon_n(\delta_2)}{\mu_c^Q - \epsilon_n(\delta_2)}$$

for cells with $\mu_c^Q > \epsilon_n(\delta_2)$. Then, with probability at least $1 - \delta_1 - \delta_2$,

$$\forall c \in \mathcal{C} : \left|\widehat{r}_c(\widehat{w}) - r_c^Q\right| \leq \underbrace{\sqrt{\frac{\log(2M/\delta_1)}{2\widehat{n_{\mathrm{eff}}}_c(\widehat{w})}}}_{\text{local label noise}} + \underbrace{\rho_c}_{\text{weight nuisance}} + \underbrace{\gamma_c(\delta_2)}_{\text{source design}}.$$

The oracle-weight case is recovered by taking $\widehat{w} = w$ and $\rho_c = 0$.

The theorem separates three sources of uncertainty. The first is the unavoidable label noise in the selected local cell, and its scale is exactly local ESS. The second is density-ratio error; cross-fitting makes the label-noise argument valid but does not make ρ_c small. The third is the difference between the observed finite source covariate design and the population target residual r_c^Q. In many audits the conditional guarantee is the most transparent report; population claims require the additional γ_c term or a sharper empirical-process analysis.

The proof has the same structure. Conditional on the covariates and fitted weights, the estimator is a weighted average of independent Bernoulli residuals. Hoeffding's inequality gives a radius controlled by the sum of squared normalized weights, which is $1/\widehat{n_{\mathrm{eff}}}_c$. The learned-weight part is a deterministic ratio perturbation: if the estimated weights differ from the oracle weights by Δ_c inside the cell, the conditional target can move by at most $2\Delta_c/(S_c - \Delta_c)$. The population term applies the same ratio perturbation to the empirical source design.

The nuisance term is best interpreted as a local robustness penalty, not as another sampling variance. It asks: how much could the cellwise residual target move if the learned density ratio is replaced by a plausible oracle ratio? When a model-based log-ratio error bound is available, the theorem gives a direct conversion. If

$$\sup_{i: A_c(X_i)=1} |\log \widehat{w}_i - \log w(X_i)| \leq \kappa_c, \qquad \kappa_c < \log 2,$$

then

$$\rho_c \le \frac{2(e^{\kappa_c}-1)}{2-e^{\kappa_c}}.$$

In most deployments, such a bound is not available from first principles. The practical substitute is a sensitivity envelope: fit a collection $\mathcal{W}$ of cross-fitted density-ratio models using different regularization levels, clipping thresholds, feature sets, and resampling perturbations, and compute

$$\widehat{\rho}_c^{\text{sens}} = \max_{\widetilde{w}\in\mathcal{W}} \frac{2\sum_i |\widehat{w}_i - \widetilde{w}_i| A_c(X_i)}{\sum_i \widetilde{w}_i A_c(X_i) - \sum_i |\widehat{w}_i - \widetilde{w}_i| A_c(X_i)},$$

over candidates with positive denominator. This is a valid ρ_c only if the candidate envelope contains the oracle ratio with high probability; otherwise it is a transparent sensitivity analysis. If $\widehat{\rho}_c^{\text{sens}}$ is comparable to the observed gap, the audit should report insufficient evidence rather than a target-certified alarm.

This also gives a useful scale calculation. Ignoring nuisance and population terms, to certify a gap $g = |\widehat{r}_c|$ above tolerance τ, one needs roughly

$$\widehat{n_{\text{eff}}}_c \gtrsim \frac{\log(2M/\delta)}{2(g-\tau)^2}.$$

For example, with $M = 150$, $\delta = 0.10$, and a margin $g - \tau = 0.20$, the required local ESS is about 100. A cell supported by ten effective labels cannot certify such a margin, no matter how large the global ESS is.

Corollary 1 (Certified alarm rule)**.** *Let*

$$B_c = \sqrt{\frac{\log(2M/\delta_1)}{2\widehat{n_{\text{eff}}}_c(\widehat{w})}} \; + \; \rho_c + \gamma_c(\delta_2).$$

If $|r_c^Q| \le \tau$ for all $c \in \mathcal{C}$, then the rule

$$\exists c \in \mathcal{C} : |\widehat{r}_c(\widehat{w})| - B_c > \tau$$

has false-alarm probability at most $\delta_1 + \delta_2$, whenever the nuisance envelope is valid.

This is the paper's practical certificate. A point estimate becomes an actionable alarm only after subtracting a simultaneous local uncertainty radius. If the point estimate exceeds τ but the certified excess is zero, the audit has not shown that the subgroup is safe; it has shown that the current evidence is insufficient.

3.3 Why Global ESS Cannot Replace Local ESS

The global ESS can be large while the selected cell is effectively supported by one weighted label.

Proposition 2 (Large global ESS with degenerate local ESS)**.** *For any target cell mass $\pi \in (0,1)$ and integer $N \ge 2$, there is a weighted sample such that the cell has weighted mass π and local ESS equal to 1, while the global ESS equals*

$$\frac{1}{\pi^2 + (1-\pi)^2/(N-1)}.$$

For $\pi = 0.05$, this quantity approaches 400 as N grows. Thus a global ESS near 400 does not rule out a worst-cell alarm supported by a single effective label.

Table 1: Synthetic audit decisions. Entries are alarm rates over 200 seeds. Under the null, every alarm is false. In the planted setting, the certified-alarm column is power.

Shift	Median global ESS	Null naive	Null global gate	Null local gate	Null certified / planted certified
0.4	625	0.99	0.99	0.38	0.00/0.98
0.8	185	1.00	0.99	0.34	0.00/0.79
1.2	47	0.99	0.46	0.08	0.00/0.45
1.6	17	0.98	0.01	0.00	0.00/0.19

3.4 Search, Stronger Estimators, and the Same Bottleneck

Theorem 1 remains valid after selecting the empirically worst cell from a pre-specified finite family $\mathcal{C}$, because the bound holds simultaneously over that family. If analysts search over a richer class after seeing labels, they must either certify over the full searched family or reserve fresh labels for the final selected cell. Sample splitting is simple: use one part of the source audit labels to discover a candidate cell, and an independent part to apply the one-cell version of the certificate.

Doubly robust estimators can reduce nuisance bias when an outcome regression is accurate, but they do not remove the local-overlap frontier. Their influence functions still contain a cell-local weighted residual term. When a target subgroup-bin cell lacks source labels, no augmentation can identify its target conditional label law without additional assumptions.

4 Experiments

The experiments are deliberately selective. Each one tests a different part of the paper's claim: local overlap can create false worst-group alarms, learned weights and adaptive search add uncertainty, and certification changes a realistic deployment decision.

All scripts and generated artifacts are in `preprint/research/`. The main studies are produced by `certified_audit_study.py`, `estimated_weight_study.py`, `adaptive_search_study.py`, `real_data_case_study.py`, and `acs_external_validation_study.py`.

4.1 Synthetic Audit: False Alarms versus Certified Failures

The first experiment uses a calibrated-by-construction covariate-shift benchmark. Source and target covariates are Gaussian with shifted means. Labels follow

$$Y \mid X \sim \text{Bernoulli}(s(X))$$

under both source and target, so every true subgroup-bin residual is zero in the null setting. Any reported worst-group failure is therefore a false alarm. In the planted-failure setting, we add a residual of size 0.60 on the interpretable subgroup $x_0 > 0.5$, which lets us measure power against a known target failure.

We compare four audit rules on the same finite subgroup-bin family: a naive worst-cell threshold, a global-ESS gate, a local-ESS gate, and the simultaneous certificate from Corollary 1. The tolerance is $\tau = 0.20$, the source audit size is $n = 1000$, and the reported study uses 200 Monte Carlo seeds with oracle Gaussian density ratios.

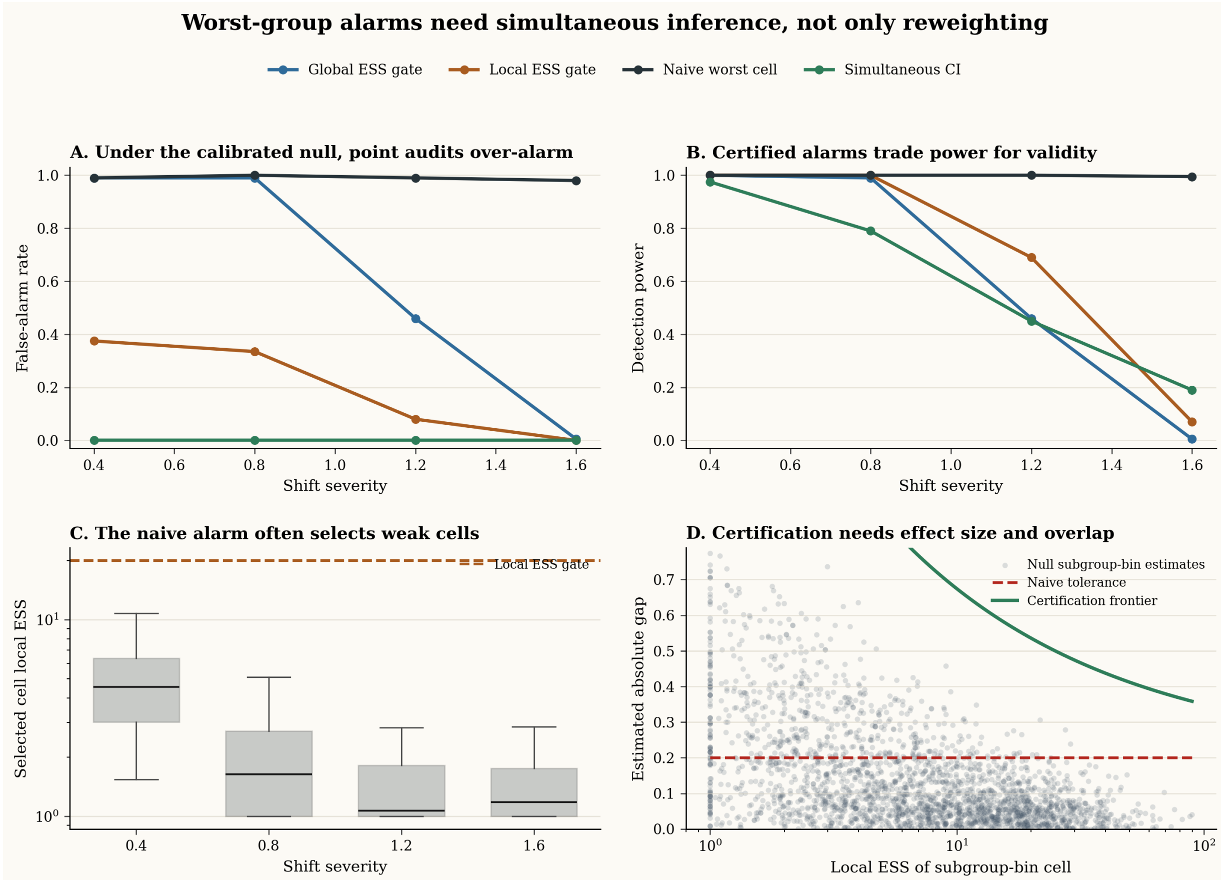

Figure 1: Synthetic covariate-shift audit. Under the calibrated null, naive and global-ESS-gated audits produce frequent false alarms, while simultaneous local certificates produce none in this run. Under a planted subgroup failure, certificates retain power when local overlap is adequate and lose power when overlap collapses. The loss of power is a feature: unsupported alarms are not certified.

Figure 1 and Table 1 show the core phenomenon. Under the null, the naive worst-cell rule false-alarms in 99%, 100%, 99%, and 98% of runs as shift increases from 0.4 to 1.6. A global ESS gate remains misleading at mild and moderate shifts, false-alarming in 99% of runs at shifts 0.4 and 0.8. A local ESS gate reduces but does not solve the problem. The simultaneous local certificate produces zero false alarms in this null experiment.

The planted-failure setting shows the other side of the tradeoff. The certified rule detects 97.5% of planted failures at shift 0.4, 79.0% at shift 0.8, 45.0% at shift 1.2, and 19.0% at shift 1.6. Certification is not a conservative decoration on the same point estimate; it changes the decision exactly when local support becomes too weak.

4.2 Learned Weights and Adaptive Search

The theorem separates two complications that are unavoidable in deployment: density ratios are estimated, and subgroups are often searched rather than fixed. We stress-test both in one experiment.

For learned weights, we compare plugin logistic density-ratio estimates with source-side cross-fitted logistic estimates. Since the synthetic density ratio is known, we can also compute the oracle cellwise

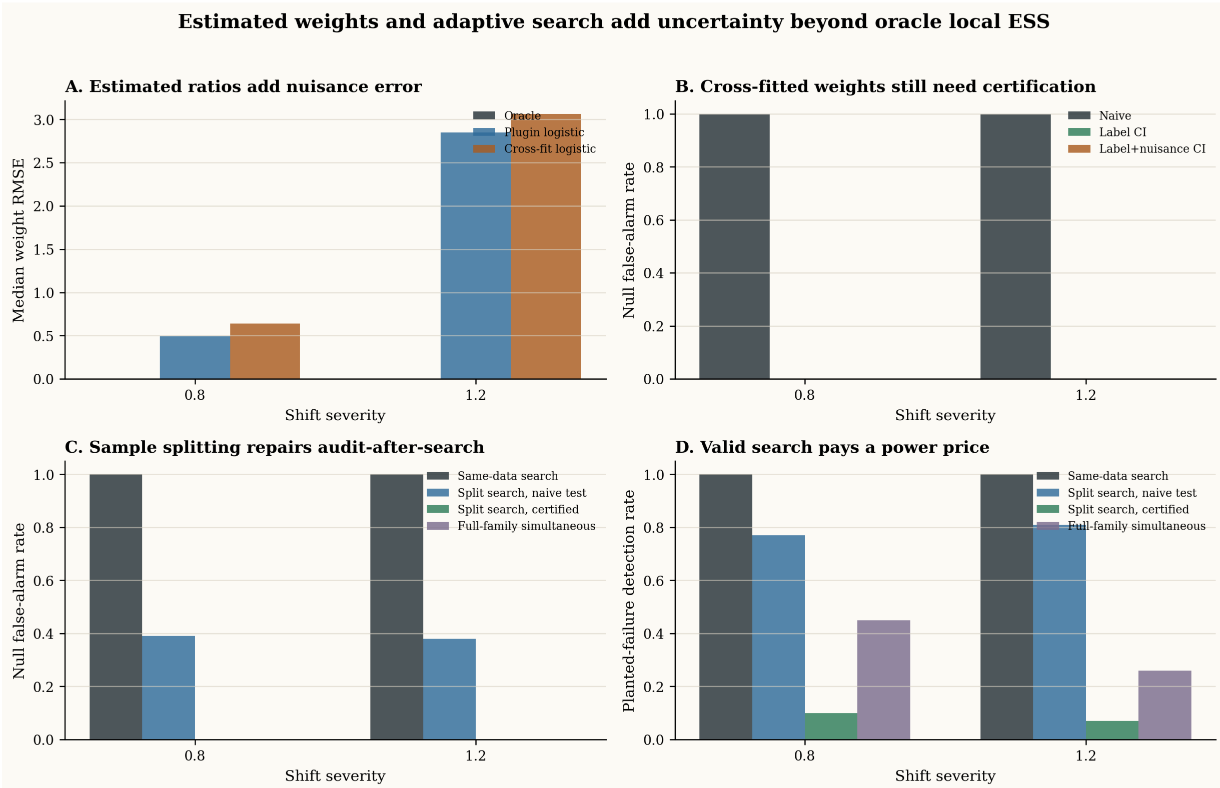

Figure 2: Estimated weights and adaptive search. Learned density-ratio weights can have reasonable global behavior while still producing large local nuisance error. Same-data adaptive search is invalid under the null; simultaneous inference or held-out certification controls false alarms but pays a power cost.

nuisance term ρ_c. This is not meant to be a deployable correction. It is a diagnostic stress test of how large the missing learned-weight term can be.

For adaptive search, we expand the subgroup class to roughly 500–600 threshold and intersection cells. We compare same-data search, full-family simultaneous inference, naive sample-split testing, and sample-split certification.

The learned-weight results match the theorem's decomposition. At shift 0.8, the median weight RMSE is 0.49 for plugin weights and 0.64 for cross-fitted weights; at shift 1.2, it rises to 2.85 and 3.06. The label-noise band with cross-fitted weights has zero null false alarms in these runs, but without a credible ρ_c envelope it certifies only the learned weighted estimand. Adding the oracle ρ_c term makes the planted-failure certificate much more conservative, detecting 1% and 0% at shifts 0.8 and 1.2. In a real audit, the oracle term would be replaced by a sensitivity envelope over plausible cross-fitted ratio models, as described in Section 5. The lesson is not that learned weights are useless. It is that local learned-weight uncertainty can dominate the alarm, and a target-certified report must show that it does not.

The adaptive-search results are equally direct. Under the calibrated null, same-data search false-alarms in 100% of runs at both tested shifts. A naive held-out threshold still false-alarms in about 39% and 38%. Full-family simultaneous inference and sample-split certification produce zero false alarms in these runs. Search is allowed, but the final alarm must pay for it.

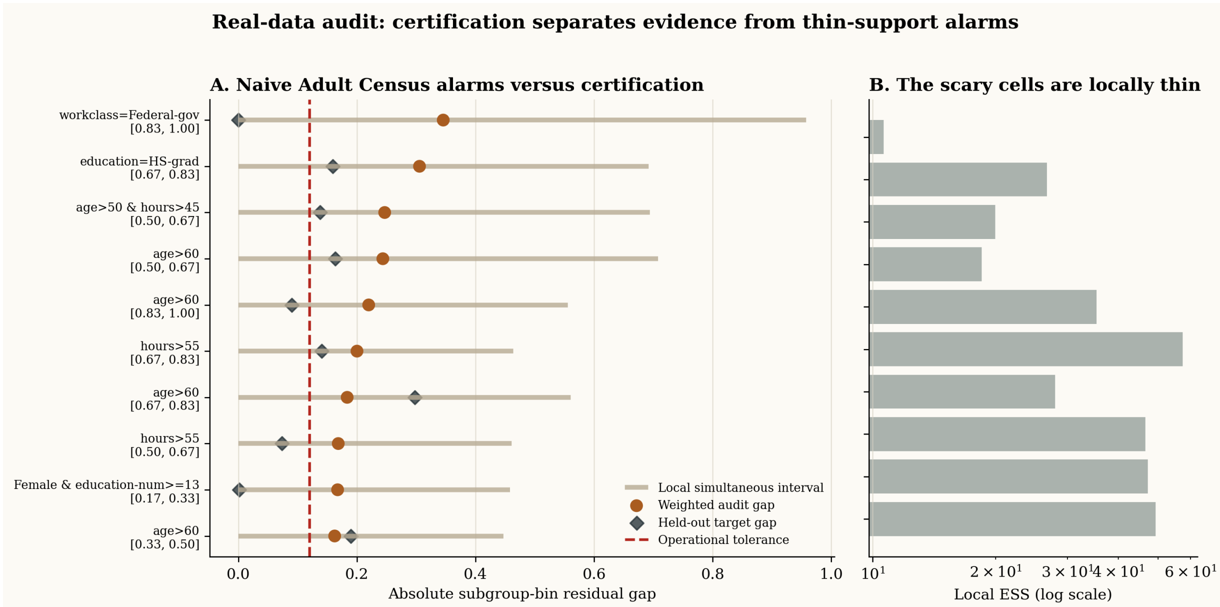

Figure 3: Adult Census deployment-style audit. The naive audit reports multiple large subgroup-bin gaps. The top alarm, `workclass=Federal-gov` in the highest score bin, has estimated gap 0.346. Its local ESS is only 10.65, its simultaneous half-width is 0.612, and the held-out target gap is 0.0005. The certified decision is insufficient evidence, not escalation.

4.3 Adult Census Deployment-Style Audit

The real-data case study asks whether the distinction matters in a plausible audit. We use the Adult Census income dataset (Kohavi, 1996). A source-trained logistic income model is audited on a target population enriched for older, higher-hours, higher-education individuals, with a mild negative tilt for the female subgroup. Because the target split depends only on covariates, it is a covariate-shift design. At audit time, the procedure uses labeled source-audit data and target covariates; target labels are held out for retrospective validation.

Figure 3 and Table 2 show the operational consequence. A naive audit would escalate the `Federal-gov` high-score subgroup. The certified audit does not claim the model is safe; it says the evidence for this alarm is too weak. Retrospective target labels support that caution for the top candidate. This is the desired behavior in deployment: avoid acting on unsupported subgroup alarms while preserving the need for more labels or coarser analysis when evidence is insufficient.

4.4 ACS Public-Service External Validation

To connect the method to a broader and more current deployment setting, we add an ACSIncome audit using Folktables (Ding et al., 2021). This is a public-service analogue of local validation: a model trained in one population is audited before use in another. We train and source-audit on California ACS records and audit a New York target population using target covariates at decision time. Target labels are used only retrospectively to interpret the audit. Cross-state external validation is not asserted to be pure covariate shift; it is a realistic stress test of how a covariate-shift audit behaves when the target subgroup dashboard looks alarming.

Figure 4 and Table 3 strengthen the empirical story in two ways. First, the pattern from Adult

Table 2: Operational interpretation of the Adult Census audit.

Audit stage	Quantity	Decision-relevant conclusion
Global readiness	global ESS = 5043, domain AUC = 0.679, 145 cells	Global diagnostics would not reject the audit.
Naive alarm	19 cells exceed $\tau = 0.12$	A point-estimate report would flag several subgroup-bin failures.
Top candidate	`workclass=Federal-gov`, score bin $[0.83, 1.00]$, $\lvert\widehat{r}_c\rvert = 0.346$	The alarm looks operationally serious.
Local certificate	local ESS = 10.65, half-width = 0.612, certified excess = 0	The correct audit decision is insufficient evidence.
Retrospective validation	held-out target gap = 0.0005 for the top candidate	The later labels support the certificate's caution.

Census is not a one-dataset artifact: global support and large naive subgroup gaps can coexist with very weak alarm-driving local support. Second, the learned-weight nuisance term matters in exactly the way the theory predicts. The top ACS alarm is already uncertified by label noise, and the sensitivity envelope makes the target-certified conclusion even less defensible. The appropriate deployment decision is again insufficient evidence, followed by more target labels or a coarser subgroup-bin analysis.

5 A Practical Audit Protocol

The methodological recommendation is compact enough to state as a protocol. A covariate-shift worst-group audit should report certified excess, not only ranked subgroup gaps.

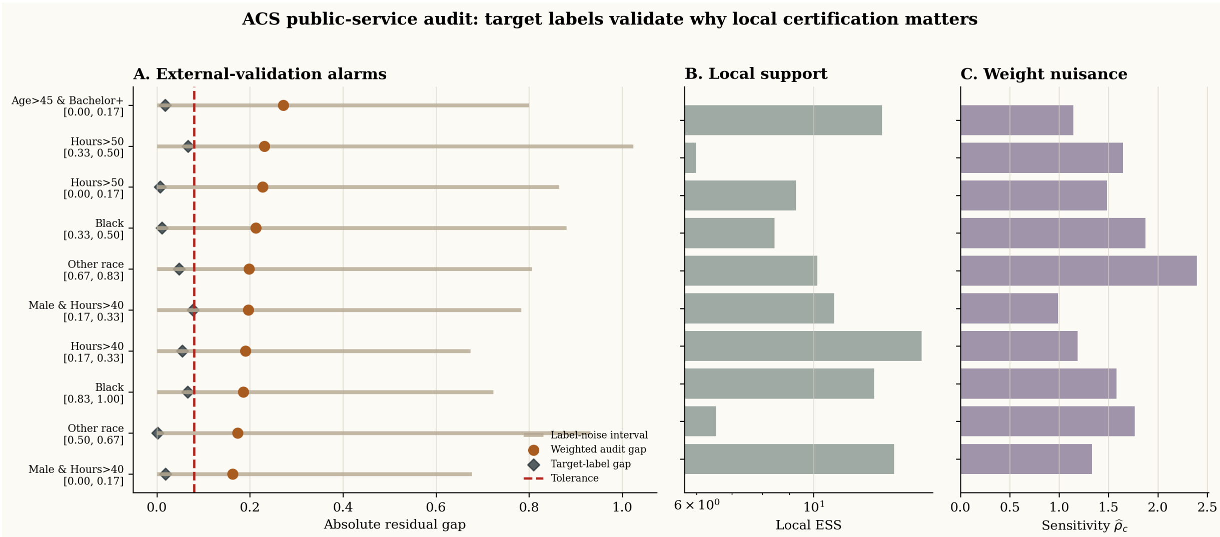

Figure 4: ACSIncome public-service external-validation audit. A California-trained income model audited for a New York target population produces many naive subgroup-bin alarms. The top alarm has estimated gap 0.272, but local ESS is 13.49, the label half-width is 0.527, and the sensitivity envelope for learned weights is $\widehat{\rho}_c = 1.143$. The retrospective target gap is only 0.018.

Certified worst-group audit under covariate shift.

1. Freeze the score, target population, tolerance τ, error budget δ, score bins, and subgroup family. If subgroups will be searched adaptively, reserve a held-out source-audit split or certify over the full searched family.

2. Estimate density ratios using source and target covariates, preferably with source-side cross-fitting. Report the clipping rule, global ESS, local ESS distribution, and any cells dominated by a few weights.

3. Build a local weight-sensitivity envelope. Fit plausible alternative ratio models by varying regularization, clipping, features, folds, and resampling. For decision-driving cells, report $\widehat{\rho}_c$ as the largest sensitivity penalty over that envelope. If the envelope is weak or unstable, label the result as sensitivity analysis rather than full target certification.

4. For every cell c, compute $\widehat{r}_c$, estimated target mass, source count, local ESS, and the simultaneous radius
$$B_c = \sqrt{\frac{\log(2M/\delta_1)}{2\widehat{n_{\mathrm{eff}}}_c}} + \widehat{\rho}_c + \widehat{\gamma}_c.$$

5. Report the certified excess
$$[|\widehat{r}_c| - \tau - B_c]_+ .$$
A cell is a certified failure only when this quantity is positive and the cell passes the pre-specified mass and support requirements.

6. Report *insufficient evidence* when the naive gap exceeds τ but certified excess is zero, local ESS is too small, $\widehat{\rho}_c$ dominates the gap, the weight-sensitivity envelope is not credible, or the subgroup was selected adaptively without valid certification.

Table 3: ACSIncome audit summary. The target labels are held out from the audit and used only to interpret the selected alarm.

Audit stage	Quantity	Decision-relevant conclusion
Global readiness	global ESS = 1055, domain AUC = 0.952, 91 cells	The target population is clearly shifted, but the weighted audit has substantial global support.
Naive alarm	30 cells exceed $\tau = 0.08$	A point-estimate report would flag many subgroup-bin failures.
Top candidate	`Age>45 & Bachelor+`, score bin $[0.00, 0.17]$, $\lvert\widehat{r}_c\rvert = 0.272$	The alarm looks operationally meaningful in a public-service audit.
Local certificate	local ESS = 13.49, label half-width = 0.527, $\widehat{\rho}_c = 1.143$	Neither the label-only nor sensitivity-adjusted certificate supports escalation.
Retrospective validation	held-out target gap = 0.018 for the top candidate	Later target labels support the insufficient-evidence decision for the top alarm.

The phrase insufficient evidence is important. It does not mean the subgroup is safe. It means the available source labels and target covariates do not support the operational claim. Appropriate next steps are to collect target labels, coarsen fragile cells, reduce the searched subgroup family, improve the density-ratio model, or defer subgroup-specific intervention.

6 Discussion

Worst-group auditing is necessary because aggregate reliability can hide real subgroup failures. Under covariate shift, however, the reported worst subgroup is also where the audit is most vulnerable to local overlap collapse and search effects. The same mechanism that makes worst-group auditing operationally useful makes it statistically delicate. This is especially relevant for local validation of clinical AI and public-service prediction systems, where subgroup alarms can determine whether a model is rolled out, restricted, recalibrated, or sent back for data collection.

The paper's central distinction is between transport and certification. Covariate-shift weighting can identify the target residual under overlap, but it does not guarantee that the selected cell has enough evidence. Global ESS measures the full weighted sample. A worst-group alarm is a local claim, so it needs a local certificate.

The learned-weight result should be read in the same spirit. Cross-fitting is valuable because it restores the conditional label-noise argument for the source labels used in the audit. It does not solve density-ratio misspecification. A journal-level treatment should therefore be explicit about the nuisance term rather than hide it behind a point estimate. In practice, $\widehat{\rho}_c$ should be reported as a local robustness number: the penalty induced by moving from the chosen ratio model to a plausible alternative ratio model in the decision-driving cell. If no credible envelope is available, the audit can still report a label-noise diagnostic, but it should not call the alarm fully target certified.

Several extensions are natural. Rich subgroup classes call for sharper complexity penalties or max-statistic calibration. Doubly robust estimators can improve constants when outcome models are useful, but they still face the same local residual variance and no-overlap boundary. Mixed shifts require assumptions beyond covariate shift. These directions are important, but they do not change the main message: a worst-group alarm under shift is a statistical claim, and it should be treated as one.

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A Proofs and Additional Experimental Details

A.1 Proof of Proposition 1

Because $P_X(A) = 0$, the labeled source sample contains no points from A. The target covariates reveal the mass $Q_X(A) = \pi$, but they reveal no labels in A. Construct two worlds that agree on P_X, Q_X, and $P(Y \mid X)$ outside A. Inside A, set

$$\mathbb{E}_{Q_0}[Y - s(X) \mid X = x] = 0$$

in world 0 and

$$\mathbb{E}_{Q_1}[Y - s(X) \mid X = x] = \Delta$$

in world 1. Both worlds satisfy covariate shift within their own joint laws because the source conditional label law can be defined to match the target conditional law on the support of P_X, and $P_X(A) = 0$. The observable distribution of source labels and target covariates is identical, while the target cell residuals differ by Δ. Therefore no procedure based on the observed data can uniformly distinguish the two cases.

A.2 Proof of Theorem 1

Fix the source and target covariates and any weight-fitting randomness. Under assumption 2, the labels are independent and the audit weights v_i are fixed. For a cell c, write

$$\alpha_{i,c} = \frac{v_i A_c(X_i)}{\sum_j v_j A_c(X_j)}.$$

Then

$$\widehat{r}_c(v) = \sum_i \alpha_{i,c} Z_i, \qquad \bar{r}_c(v) = \sum_i \alpha_{i,c} m_i, \qquad \sum_i \alpha_{i,c}^2 = \frac{1}{\widehat{n_{\text{eff}}}_c(v)}.$$

Conditional on X_i, the Bernoulli residual $Z_i = Y_i - s(X_i)$ takes values in an interval of length one. Hoeffding's inequality for weighted sums gives

$$\mathbb{P}\left(|\widehat{r}_c(v) - \bar{r}_c(v)| > t\right) \leq 2\exp\{-2t^2 \widehat{n_{\text{eff}}}_c(v)\}.$$

Taking

$$t = \sqrt{\frac{\log(2M/\delta_1)}{2\widehat{n_{\text{eff}}}_c(v)}}$$

and applying a union bound over M cells proves the first part.

For learned weights, let

$$\widehat{S}_c = \sum_i \widehat{w}_i A_c(X_i), \qquad S_c = \sum_i w_i A_c(X_i), \qquad N_c = \sum_i w_i A_c(X_i) m_i.$$

Because $|m_i| \leq 1$,

$$\left|\sum_i (\widehat{w}_i - w_i) A_c(X_i) m_i\right| \leq \Delta_c, \qquad |\widehat{S}_c - S_c| \leq \Delta_c, \qquad \widehat{S}_c \geq S_c - \Delta_c.$$

The elementary ratio inequality

$$\left|\frac{\widehat{N}}{\widehat{S}} - \frac{N}{S}\right| \leq \frac{|\widehat{N} - N|}{\widehat{S}} + \frac{|N|\,|\widehat{S} - S|}{S\widehat{S}}$$

with $|N| \leq S$ yields

$$|\bar{r}_c(\widehat{w}) - \bar{r}_c(w)| \leq \frac{2\Delta_c}{S_c - \Delta_c} \leq \rho_c.$$

Adding this deterministic perturbation to the label-noise event proves the learned-weight statement.

For the population term, assume $0 \leq w(X) \leq W$. The variables

$$w(X_i) A_c(X_i) m_i \quad \text{and} \quad w(X_i) A_c(X_i)$$

both lie in intervals of length at most $2W$ and W, respectively, and the first has absolute value at most W. A finite-family Hoeffding bound gives, with probability at least $1 - \delta_2$,

$$\left| \frac{1}{n} \sum_i w(X_i) A_c(X_i) m_i - \mathbb{E}_P[w(X) A_c(X) m(X)] \right| \le \epsilon_n(\delta_2)$$

and

$$\left| \frac{1}{n} \sum_i w(X_i) A_c(X_i) - \mu_c^Q \right| \le \epsilon_n(\delta_2)$$

simultaneously over cells, after adjusting constants. On this event, the same ratio perturbation argument gives

$$|\bar{r}_c(w) - r_c^Q| \le \frac{2\epsilon_n(\delta_2)}{\mu_c^Q - \epsilon_n(\delta_2)} = \gamma_c(\delta_2).$$

Combining the label, learned-weight, and population events gives probability at least $1 - \delta_1 - \delta_2$.

A.3 Proof of Corollary 1

On the event from Theorem 1,

$$|\widehat{r}_c(\widehat{w}) - r_c^Q| \le B_c$$

simultaneously over c. If $|r_c^Q| \le \tau$ for every cell, then

$$|\widehat{r}_c(\widehat{w})| - B_c \le |r_c^Q| \le \tau$$

for every cell. Therefore no certified alarm occurs on that event. The failure probability is at most $\delta_1 + \delta_2$.

A.4 Proof of Proposition 2

Take one active point, $A_1 = 1$, and $N - 1$ inactive points, $A_2 = \cdots = A_N = 0$. Assign

$$w_1 = N\pi, \qquad w_2 = \cdots = w_N = \frac{N(1-\pi)}{N-1}.$$

The weighted mass of the active cell is π, and its local ESS is 1. The global ESS is

$$\frac{(N\pi + N(1-\pi))^2}{(N\pi)^2 + (N-1)\{N(1-\pi)/(N-1)\}^2} = \frac{1}{\pi^2 + (1-\pi)^2/(N-1)}.$$

A.5 Additional Experimental Details

The synthetic certification experiment uses oracle density-ratio weights for the Gaussian shift benchmark. The null setting is calibrated by construction. The planted setting adds a residual shift of 0.60 on $x_0 > 0.5$, clipped to keep probabilities in $[10^{-3}, 1 - 10^{-3}]$. The four rules are the naive worst-cell threshold, a global ESS gate at 50, a local ESS gate at 20, and the simultaneous local certificate.

A separate local-overlap diagnostic run, reported in earlier drafts and preserved in the repository, shows that lower-tail local ESS collapses much faster than global ESS. At shift 0.8, the median global ESS is about 95, the lower-tail local ESS is about 1.4, and the false-alarm rate above 0.05 is about 80%.

The learned-weight experiment compares plugin logistic density-ratio weights and source-side cross-fitted logistic weights. Because the oracle ratio is known in this benchmark, we compute the nuisance term

$$\frac{2\Delta_c}{S_c - \Delta_c}$$

whenever $S_c > \Delta_c$, and set it to 1 otherwise. This deliberately conservative oracle correction is used only to show how local density-ratio error can dominate a certificate.

A.6 Constructing $\widehat{\rho}_c$ in Practice

The population quantity ρ_c depends on the unknown oracle ratio. A deployable audit should therefore distinguish three levels of evidence.

Model-based certificate. If the density-ratio procedure comes with a high-probability local log-error bound

$$|\log \widehat{w}_i - \log w(X_i)| \le \kappa_c \qquad \text{for all } i \text{ with } A_c(X_i) = 1,$$

then the main text gives the plug-in envelope

$$\widehat{\rho}_c = \frac{2(e^{\kappa_c} - 1)}{2 - e^{\kappa_c}}, \qquad \kappa_c < \log 2.$$

Such a bound might come from a correctly specified parametric ratio model, a conservative uncertainty band for the domain classifier's logit, or an externally validated density-ratio model. If $\kappa_c \ge \log 2$, the bound is too weak to certify the cell.

Sensitivity envelope. More commonly, the audit team has no theorem-level guarantee for the ratio model. The practical alternative is to construct a candidate set $\mathcal{W}$ of plausible cross-fitted ratios. Useful perturbations include:

- varying the domain-classifier class or regularization strength;
- varying clipping thresholds for extreme weights;
- refitting over bootstrap or subsampled source/target covariates;
- adding or removing covariate blocks whose shift mechanism is uncertain;
- comparing direct density-ratio methods with probabilistic domain-classifier ratios.

For a baseline $\widehat{w}$, compute

$$\widehat{\rho}_c^{\text{sens}} = \max_{\widetilde{w} \in \mathcal{W}} \frac{2\sum_i |\widehat{w}_i - \widetilde{w}_i| A_c(X_i)}{\sum_i \widetilde{w}_i A_c(X_i) - \sum_i |\widehat{w}_i - \widetilde{w}_i| A_c(X_i)},$$

omitting candidates with nonpositive denominators and marking the cell unsupported if all denominators fail. This quantity is a rigorous nuisance term only under the additional assumption that $\mathcal{W}$ contains the oracle ratio with high probability. Without that assumption, it is still valuable as a transparent stress test: it tells the reader how much the cell conclusion changes across plausible ratio models.

Balance diagnostics. Weight sensitivity should be accompanied by covariate balance checks on held-out source and target covariates. For each decision-driving cell, compare the target cell mass

$$\widehat{\mu}_c^Q = \frac{1}{m}\sum_{j=1}^{m} A_c(X_j')$$

with the weighted source estimate

$$\widehat{\mu}_c^{P,\widehat{w}} = \frac{1}{n}\sum_{i=1}^{n} \widehat{w}_i A_c(X_i),$$

and repeat the comparison for important covariate summaries inside the cell. Large imbalance does not directly equal ρ_c, but it is evidence that the ratio model is not transporting the local cell reliably. The appropriate report is then insufficient evidence unless the alarm survives a wider sensitivity envelope.

In all cases, $\widehat{\rho}_c$ should be reported only for cells that could affect the decision: the selected worst cell, any certified cells, and a small set of near-threshold cells. Reporting a single global nuisance number hides the same local problem that motivates the paper.

The adaptive-search experiment expands the subgroup class to threshold rules on all six synthetic features and pairwise positive-tail intersections over the first four features, producing roughly 500–600 nonempty cells after filtering. The sample-split protocol uses half of the source audit labels for discovery and half for the final one-cell certificate.

The Adult Census case study uses `fetch_openml("adult", version=2)` and removes `fnlwgt`. The target deployment split is sampled with probability increasing in age, hours worked, and college-level education, with a mild negative tilt for the female subgroup. A logistic income model is trained on 16000 source-training examples. The audit uses 9000 labeled source-audit examples, 12000 target covariates, and cross-fitted logistic domain weights. Target labels are used only for retrospective evaluation.

The ACSIncome case study uses Folktables (Ding et al., 2021). The source population is California ACS records and the target population is New York ACS records from the 2018 one-year person survey. A logistic income model is trained on 30000 California records and audited on 18000 held-out California source-audit records against 18000 New York target covariates. The subgroup family uses sex, race, age, hours worked, education, and simple intersections. The target labels are held out from the audit and used only for retrospective validation. The learned-weight sensitivity envelope is built from alternative cross-fitted domain-ratio models with different regularization strengths and clipping thresholds.

A.7 Secondary Results Kept Out of the Main Text

The main text focuses on certification, but the repository also contains subgroup-degradation examples showing why worst-group auditing is needed in the first place. In representative synthetic scenarios, masked subgroup shift gives a worst-group ECE gap of about 0.182, and minority subgroup degradation gives a gap of about 0.217. These examples are not used as headline evidence because the paper's contribution is not the existence of hidden subgroup failures; it is the certification problem under covariate shift.

The same reason explains why doubly robust estimators are discussed but not foregrounded. For a residual $Z = Y - s(X)$, a typical augmented residual-mass estimator has the form

$$\widehat{\theta}_c^{\mathrm{DR}} = \frac{1}{m}\sum_{j=1}^{m}\widehat{m}(X_j')A_c(X_j') + \frac{1}{n}\sum_{i=1}^{n}\widehat{w}_i A_c(X_i)\{Z_i - \widehat{m}(X_i)\}.$$

If $\widehat{m}$ is accurate, this can reduce nuisance bias and variance. However, the influence-function variance still contains a local weighted residual term,

$$\mathbb{E}_P\Big[w(X)^2 A_c(X)\,\mathrm{Var}(Z \mid X)\Big]\,,$$

and the no-overlap result remains unchanged. Augmentation can improve constants; it cannot identify labels in an unsupported target cell.

The main rendered artifacts are:

- `preprint/figures/certified_audit_main.png`
- `preprint/figures/estimated_weight_and_search.png`
- `preprint/figures/adult_case_study.png`
- `preprint/figures/acs_external_validation.png`
- `preprint/research/results/certified_audit/`
- `preprint/research/results/estimated_weights/`
- `preprint/research/results/adaptive_search/`
- `preprint/research/results/adult_case_study/`
- `preprint/research/results/acs_external_validation/`